SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA5112

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: Implement best security practices for IP, email, and web service using PGP,S/MIME for enhancing email Security.CO (BL3, BL6)

Assessment Tool Weightage: 20%

QUESTIONS: 5 TOTAL MARKS: 25

Annexure A

Industry Problem-Solving Task

Code of Conduct:

I E .Lakshmi Narayanan(192524249) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

E .Lakshmi Narayanan

1. SSL/TLS Overhead and Web Server Performance Optimization

Given:

- HTTP payload per response = **2000 bytes**
- TLS overhead per response = **25 bytes**
- Handshake overhead = **3 KB/session**
- Requests per second = **10,000**
- New TLS handshakes per second = **1,000**
- Assume **1 KB = 1024 bytes**

A. Total data transmitted per second

TLS-protected response size:

$2000+25=2025$ bytes

For 10,000 requests:

$2025\times10000=20{,}250{,}000$ bytes/s

Handshake data:

$3\times1024=3072$ bytes

For 1,000 handshakes:

$3072\times1000=3{,}072{,}000$ bytes/s

Therefore:

$20{,}250{,}000+3{,}072{,}000=23{,}322{,}000$ bytes/s

Approximately:

23.32 MB/s

In bits:

$23{,}322{,}000 \times 8 = 186.576$ Mbps

B. Percentage overhead due to TLS encryption

TLS record overhead:

$200025 \times 100 = 1.25\%$

Including handshake overhead, total TLS overhead is:

$25 \times 10000 + 3{,}072{,}000 = 3{,}322{,}000$ bytes/s

Original application data:

$2000 \times 10000 = 20{,}000{,}000$ bytes/s

Overall overhead percentage:

$20{,}000{,}0003{,}322{,}000 \times 100 = 16.61\%$

C. Bandwidth consumed by handshake overhead

$3072 \times 1000 = 3{,}072{,}000$ bytes/s

In Mbps:

$3{,}072{,}000 \times 8 = 24{,}576{,}000$ bits/s 24.576 Mbps

D. Trade-off and optimization

TLS provides **confidentiality, integrity, and authentication**, but encryption and handshakes introduce CPU, bandwidth, and latency overhead.

Optimization techniques:

1. **TLS session resumption** – avoids a full handshake for returning clients.
2. **HTTP/2** – multiplexes multiple requests over one TLS connection, reducing connection overhead.
3. **QUIC/HTTP/3** – reduces connection establishment latency and handles packet loss efficiently.
4. **Persistent connections** – reuse existing TLS connections instead of creating new ones.
5. **Hardware acceleration** – cryptographic hardware can reduce encryption CPU usage.

Conclusion: Security should be retained, while session reuse and modern protocols can significantly reduce TLS latency and overhead.

2. PGP-Based Email Encryption Load in Enterprise System

Given:

- Attachment size = **10 MB**
- PGP overhead = **800 bytes/email**
- Emails per day = **5,000**
- Assume 1 MB = 1,048,576 bytes.

A. Total data transmitted per day

10 MB in bytes:

$10 \times 1{,}048{,}576 = 10{,}485{,}760$ bytes

Size after PGP:

$10{,}485{,}760 + 800 = 10{,}486{,}560$ bytes

For 5,000 emails:

$10{,}486{,}560 \times 5000 = 52{,}432{,}800{,}000$ bytes/day

Approximately:

52.43 GB/day

B. Percentage increase in storage

$10{,}485{,}760800 \times 100 = 0.00763\%$

Thus, PGP adds only about **0.0076%** storage overhead.

C. Computational overhead

PGP generally uses a **hybrid encryption approach**:

- A symmetric algorithm encrypts the large email/attachment.
- Public-key cryptography protects the symmetric session key.
- Digital signatures provide authentication and integrity.
- Decryption reverses these operations.

For 10 MB attachments, the **symmetric encryption/decryption of the data** is the major data-processing task, while public-key operations add computational overhead for key handling and signatures.

D. Optimization strategies

1. **Compression before encryption** – reduces the amount of data encrypted and transmitted.
2. **Batch processing** – process multiple emails efficiently during high-load periods.
3. **Hardware acceleration** – cryptographic hardware can improve encryption performance.
4. **Efficient key management** – avoid unnecessary public-key operations.
5. **Parallel processing** – encrypt multiple emails simultaneously using multiple CPU cores.

Conclusion: PGP's storage overhead is extremely small, but encryption/decryption can consume significant CPU resources when thousands of large attachments are processed.

3. IPSec Transport vs Tunnel Mode Performance Comparison

Given:

- Payload = **1000 bytes**
- Original header = **20 bytes**
- Transport overhead = **30 bytes**
- Tunnel overhead = **50 bytes**
- Traffic = **20,000 packets/second**

A. Total packet size

Original packet:

$1000+20=1020$ bytes

Transport mode

$1020+30=1050$ bytes

Tunnel mode

$1020+50=1070$ bytes

B. Bandwidth usage

Transport mode

$1050\times 20,000=21,000,000$ bytes/s

In Mbps:

$21,000,000 \times 8 = 168$ Mbps

Tunnel mode

$1070 \times 20,000 = 21,400,000$ bytes/s

In Mbps:

$21,400,000 \times 8 = 171.2$ Mbps

Comparison

Mode	Packet Size	Bandwidth
Transport	1050 bytes	168 Mbps
Tunnel	1070 bytes	171.2 Mbps

Tunnel mode uses:

$171.2 - 168 = 3.2$ Mbps

more bandwidth.

C. Efficiency and security

Transport mode:

- Smaller overhead.
- Better bandwidth efficiency.
- Protects the payload of the original IP packet.
- Suitable when communicating directly between trusted hosts.

Tunnel mode:

- Higher overhead.
- Encapsulates the entire original IP packet.
- Provides stronger protection for the original packet structure.
- Commonly used for VPNs and gateway-to-gateway communication.

D. Recommendation

Internal communication:
Use **Transport Mode** when endpoints are trusted and only the payload needs protection. It has lower overhead and better efficiency.

External communication:
Use **Tunnel Mode**, especially for VPN communication across untrusted networks, because it protects the complete original IP packet and hides internal addressing information.

4. S/MIME Certificate Management in Large Organization

Given:

- Employees = **12,000**
- Certificate size = **2 KB**
- During renewal, old + new certificates are stored.
- Renewal period = **1 month**

A. Total storage during renewal

Old certificates:

$12{,}000 \times 2\ \text{KB} = 24{,}000\ \text{KB}$

New certificates:

$12{,}000 \times 2\ \text{KB} = 24{,}000\ \text{KB}$

Total:

$24{,}000 + 24{,}000 = 48{,}000\ \text{KB}$

Approximately:

48 MB

or approximately **46.875 MiB** if using 1024-based units.

B. Certificate distribution overhead

If each new 2 KB certificate is distributed to all 12,000 employees:

$12{,}000 \times 2\ \text{KB} = 24{,}000\ \text{KB}$

So approximately:

24 MB

of certificate data must be distributed during renewal.

The actual network overhead will be slightly higher because of protocols, acknowledgements, authentication, retries, and other network headers.

C. Challenges at scale

Managing 12,000 certificates creates several challenges:

1. **Certificate expiration** – expired certificates can prevent secure email communication.
2. **Renewal management** – thousands of certificates must be renewed correctly.
3. **Revocation** – compromised certificates need to be revoked quickly.
4. **Distribution** – certificates must reach the correct users/devices.
5. **Key management** – private keys must be protected.
6. **Monitoring** – administrators need to track certificate status.
7. **Human error** – manual certificate management can cause configuration mistakes.

D. Optimization strategies

- **Automated certificate renewal** reduces manual work.
- **Centralized PKI** provides consistent certificate management.

- **Automated deployment** distributes certificates to employee devices.
- **Certificate monitoring** provides alerts before expiration.
- **Short-lived certificates** can reduce the impact of compromised credentials.
- **Centralized revocation management** allows faster response to security incidents.

Conclusion: The storage requirement is relatively small, but managing thousands of certificates is operationally complex. Automation and centralized PKI are the most effective solutions.

5. Email Encryption Latency in Cloud-Based Systems

Given:

- AES encryption = **0.2 ms/MB**
- RSA key exchange = **5 ms/email**
- Attachment = **8 MB**
- Emails = **8,000/hour**

A. Total encryption time per email

AES time:

$8 \times 0.2 = 1.6$ ms

RSA time:

5 ms

Total:

$1.6 + 5 = 6.6$ ms/email

B. Total processing time per hour

$6.6 \times 8000 = 52{,}800$ ms

Convert to seconds:

$52{,}800 / 1000 = 52.8$ seconds/hour

So the total encryption computation is approximately:

52.8 seconds per hour

assuming the operations are processed sequentially.

C. Performance bottleneck

AES:

1.6 ms

RSA:

5 ms

Total:

6.6 ms

Percentage contributed by RSA:

$6.65 \times 100 = 75.76\%$

Percentage contributed by AES:

$6.61.6 \times 100 = 24.24\%$

Therefore, **RSA is the major performance bottleneck**.

RSA takes more time per email because asymmetric cryptographic operations are computationally more expensive than symmetric AES encryption.

D. Optimization strategies

1. **ECC adoption** – ECC can provide strong security with smaller keys and generally lower computational/network overhead than RSA in many key-establishment scenarios.
2. **Parallel processing** – process multiple emails simultaneously using multiple CPU cores.
3. **Hardware acceleration** – use AES-NI or cryptographic accelerators to speed up symmetric encryption.
4. **Session/key reuse where appropriate** – avoid unnecessary repeated asymmetric operations.
5. **Asynchronous processing** – move encryption into background workers so that email delivery isn't blocked unnecessarily.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation